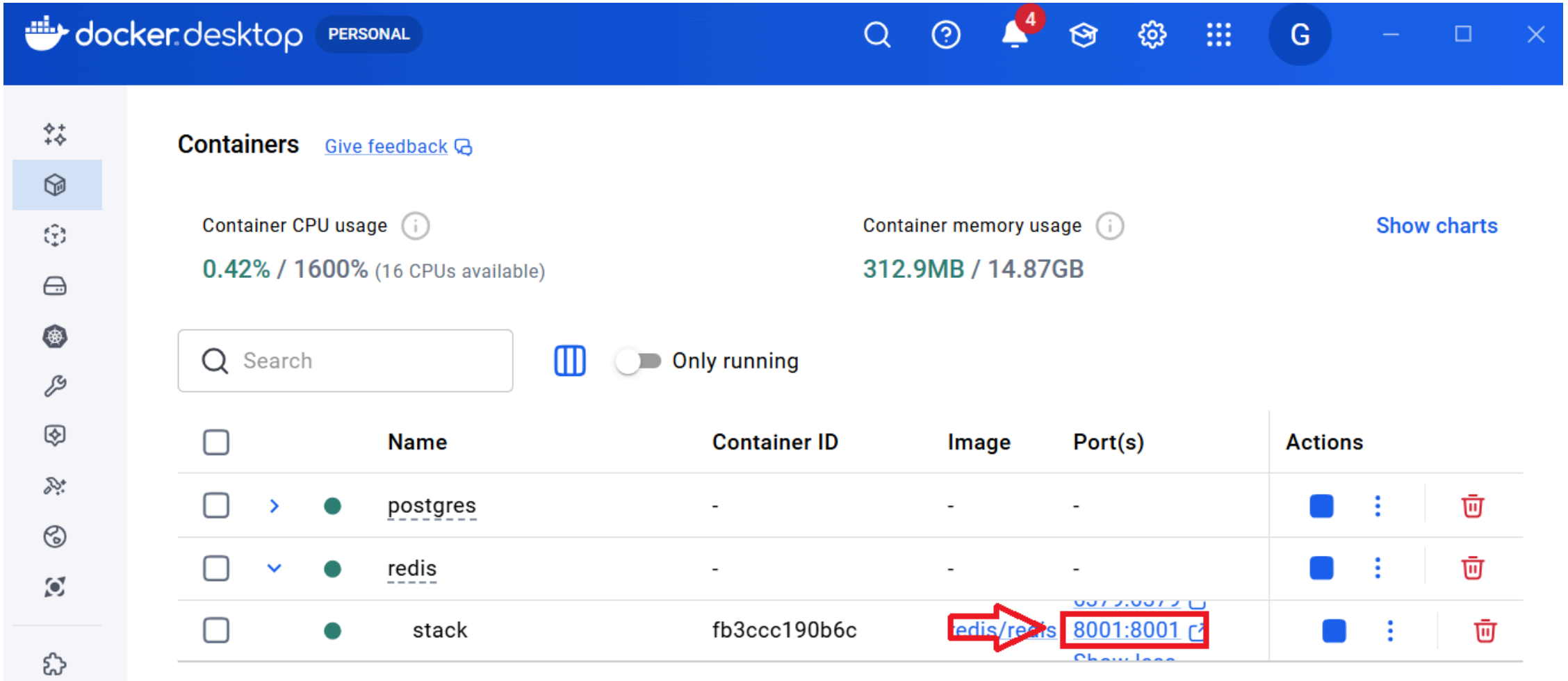
키와 기본 명령어

Redis UI 접속


docker desktop
PERSONAL
4
G
Containers
Give feedback
Container CPU usage
0.42% / 1600% (16 CPUs available)
Container memory usage
312.9MB / 14.87GB
Show charts
Search
Only running
Name
Container ID
Image
Port(s)
Actions
postgres
-
-
-
redis
-
-
-
stack
fb3ccc190b6c
8001:8001

1. 키 이름 설계

Redis에는 테이블이 없으므로 키 이름이 데이터의 의미를 전달해야 합니다.
이 강의에서는 다음 규칙을 사용합니다.

```
서비스:대상:식별자[:속성]
```

```
shop:product:1001
shop:product:1001:views
community:user:7:session
```

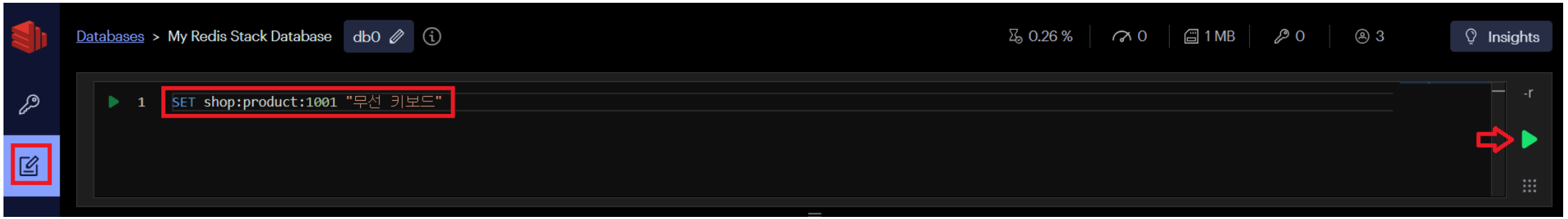
좋은 키의 조건

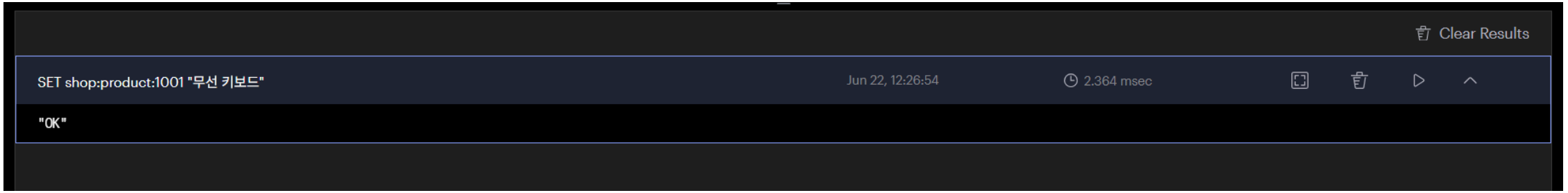
- 의미가 분명하고 충돌하지 않습니다.
- 너무 짧거나 지나치게 길지 않습니다.
- 같은 종류의 키는 같은 패턴을 사용합니다.
- 개인정보를 키 이름에 직접 넣지 않습니다.

2. 기본 저장과 조회

- 키(key): `shop:product:1001`
- 데이터(value): `무선 키보드`

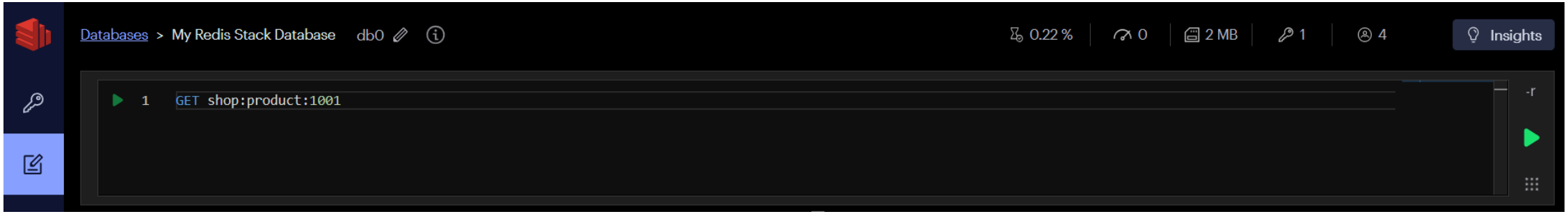
```
SET shop:product:1001 "무선 키보드"
```



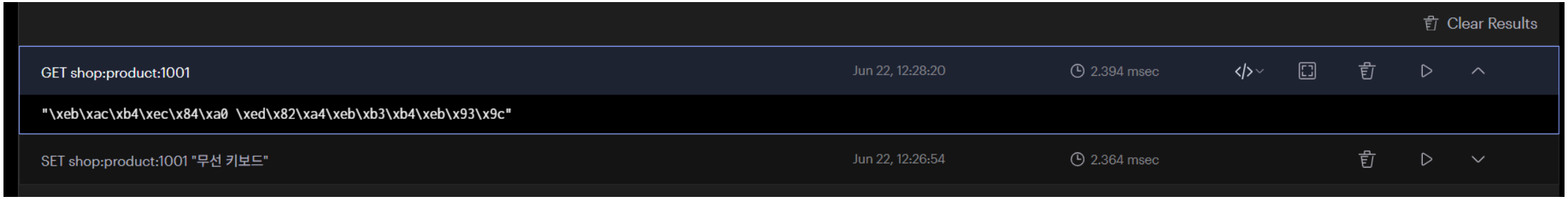


- 키(key): `shop:product:1001`

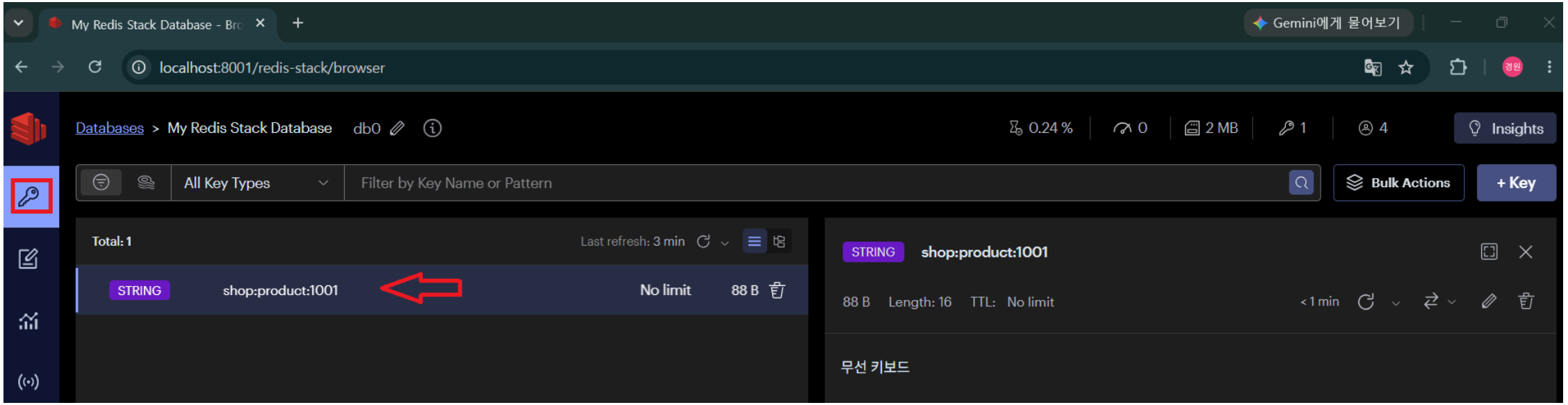
```
GET shop:product:1001
```



> 한글이 깨진 게 아니라 Redis가 문자열을 bytes escape 형태로 보여주는 것입니다.

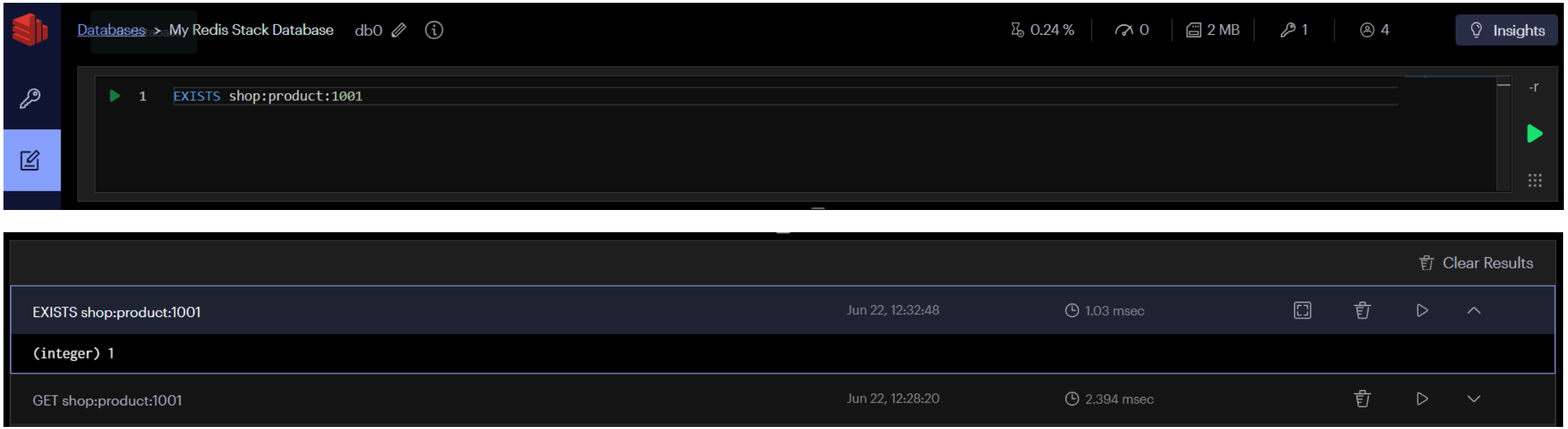


browser에서 데이터 확인



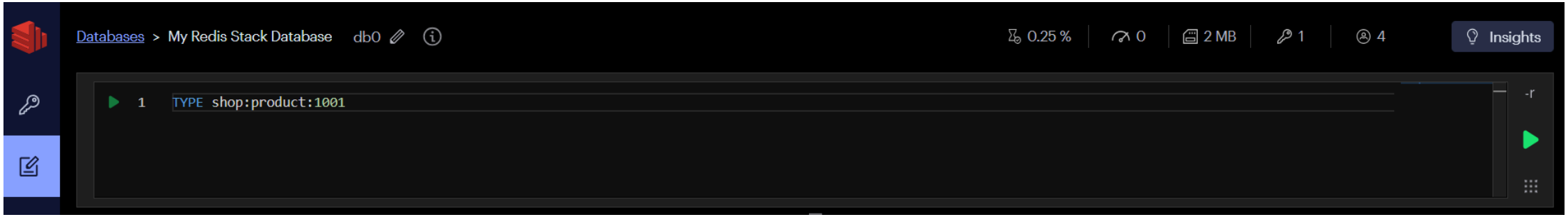
- `EXISTS key` : 키가 있으면 `1`, 없으면 `0`

```
EXISTS shop:product:1001
```



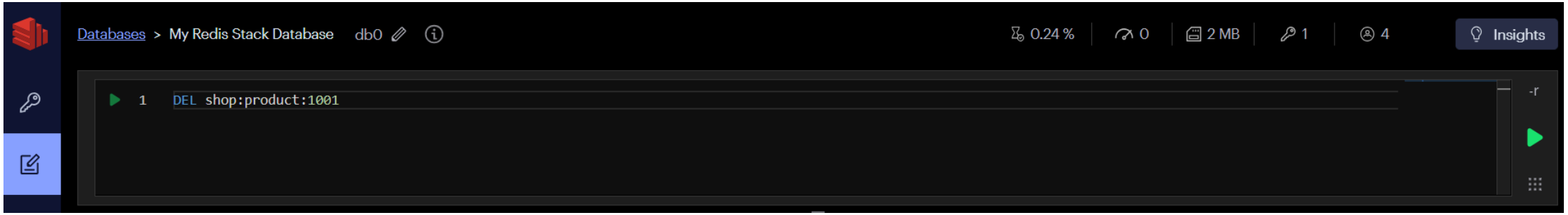
- `TYPE key` : 키에 저장된 자료형 확인

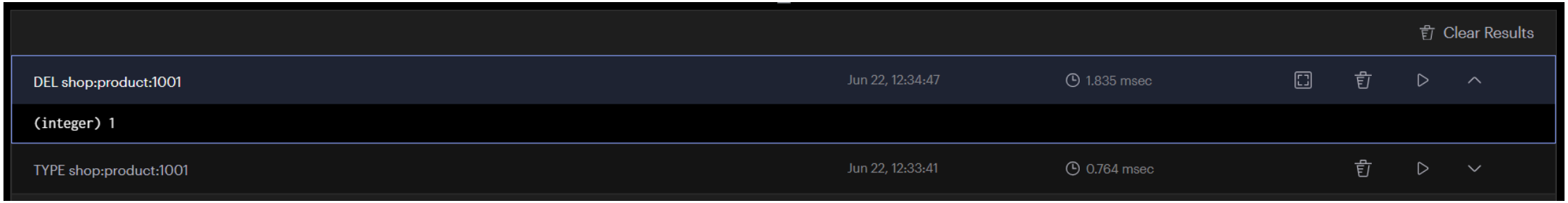
```
TYPE shop:product:1001
```



- `DEL key [key ...]` : 하나 이상의 키를 즉시 삭제

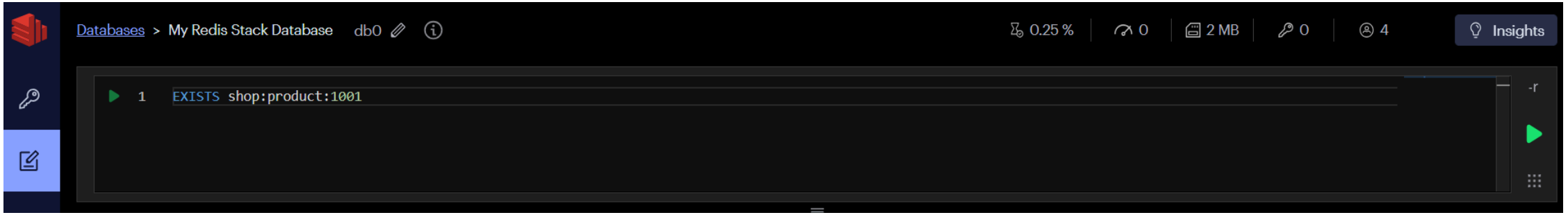
```
DEL shop:product:1001
```





- `EXISTS key` : 키가 있으면 `1`, 없으면 `0`

```
EXISTS shop:product:1001
```

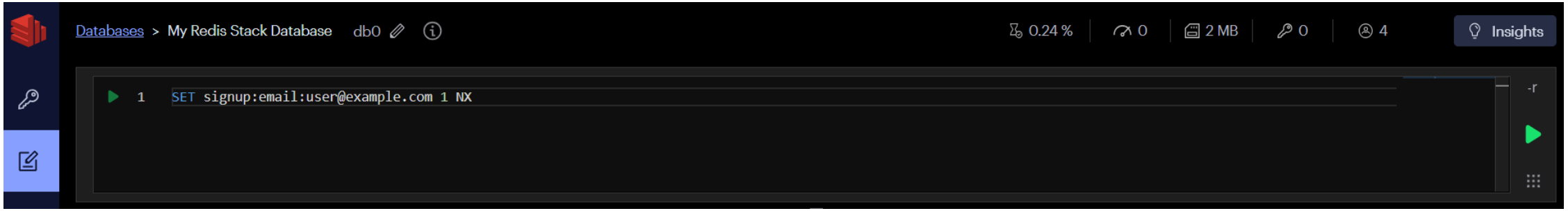


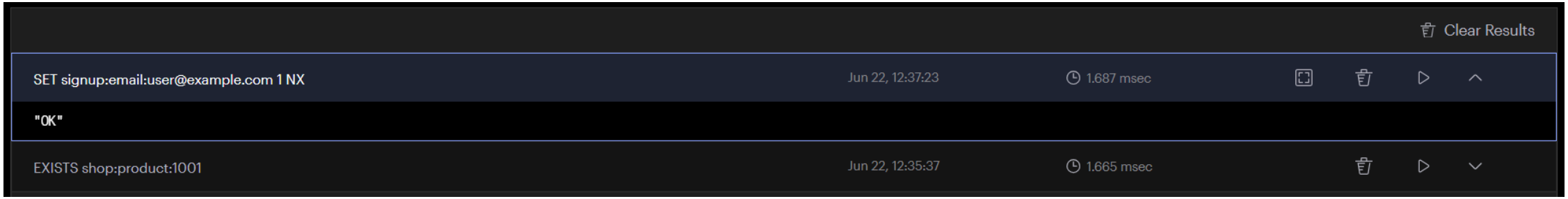
3. SET 옵션

> 조건을 만족하지 못한 `SET`은 `(nil)`을 반환합니다. `NX`는 간단한 중복 방지에 유용하지만, 복잡한 분산 잠금은 만료와 소유권 확인까지 별도로 설계해야 합니다.

- `NX` : 키가 없을 때만 저장

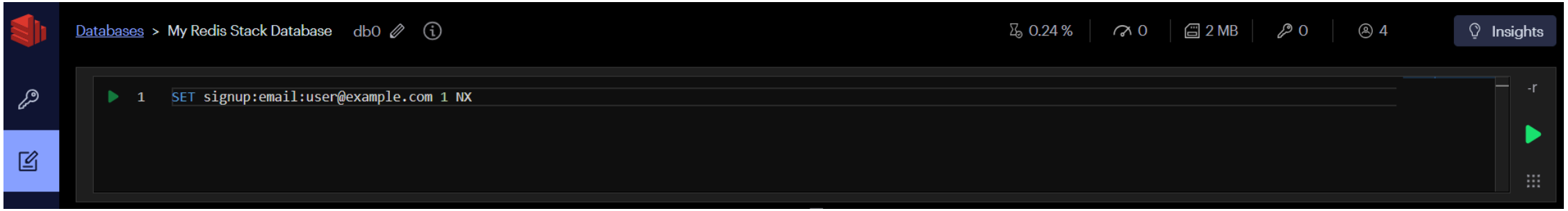
```
SET signup:email:user@example.com 1 NX
```



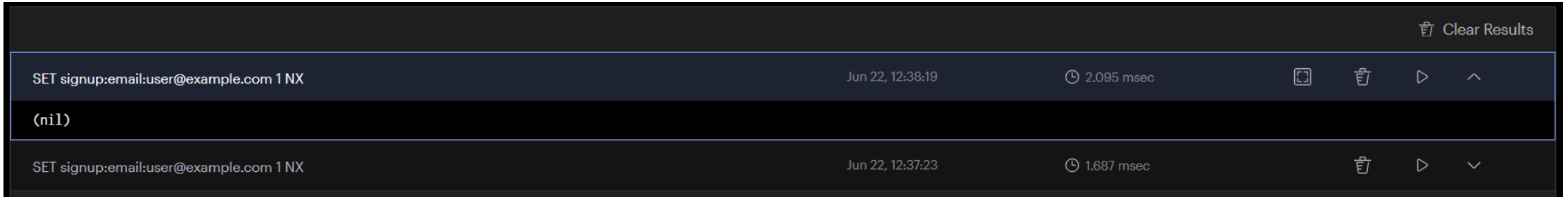


- `NX` : 키가 없을 때만 저장

```
SET signup:email:user@example.com 1 NX
```

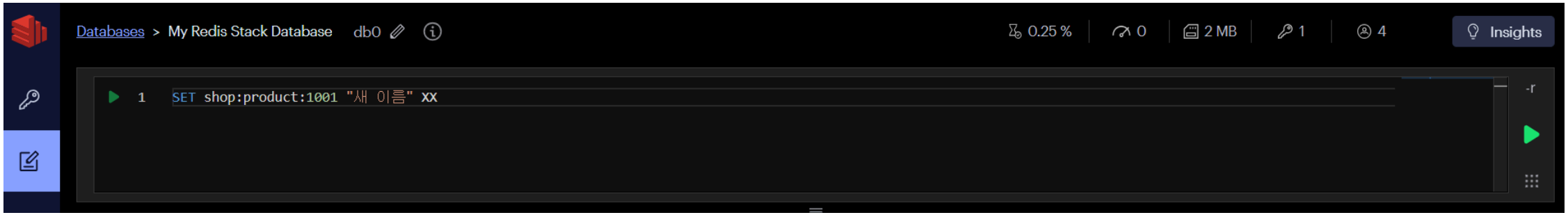


> 이미 저장된 키(signup:email:user@example.com)가 있기 때문에 저장 실패



- `XX` : 키가 있을 때만 저장

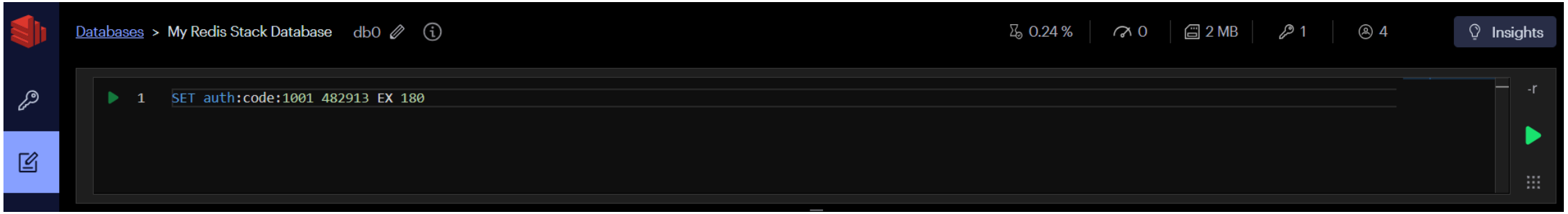
```
SET shop:product:1001 "새 이름" XX
```

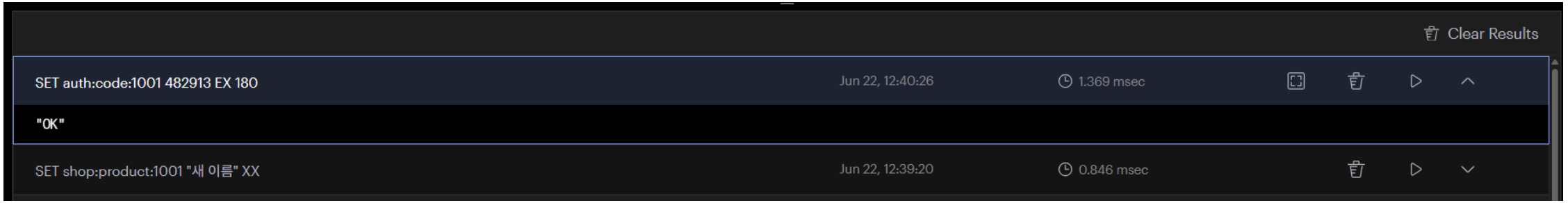


> 저장된 키(shop:product:1001)가 없기 때문에 저장 실패

- `EX 초` : 저장과 동시에 초 단위 만료 시간 설정
- `PX 밀리초` : 저장과 동시에 밀리초 단위 만료 시간 설정

```
SET auth:code:1001 482913 EX 180
```



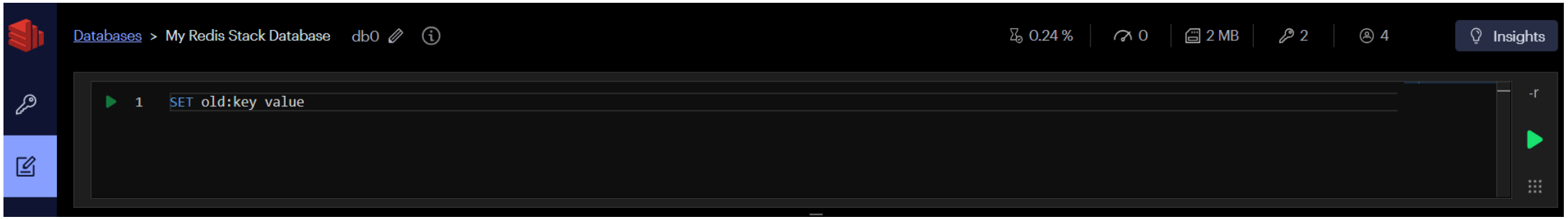


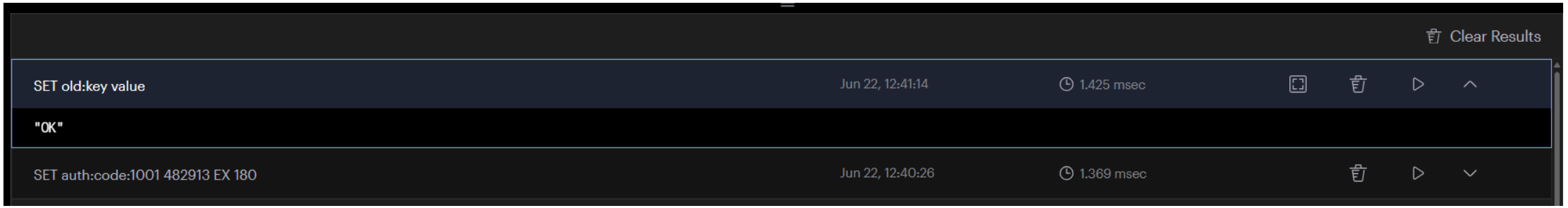
4. 키 이름 변경

> `RENAME`은 대상 키가 이미 있으면 덮어씁니다. 덮어쓰기를 막으려면 `RENAMENX`를 사용합니다.

- 키 생성

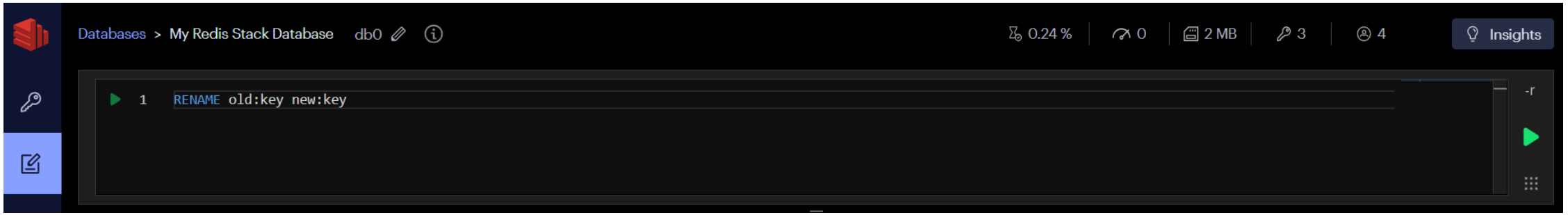
```
SET old:key value
```





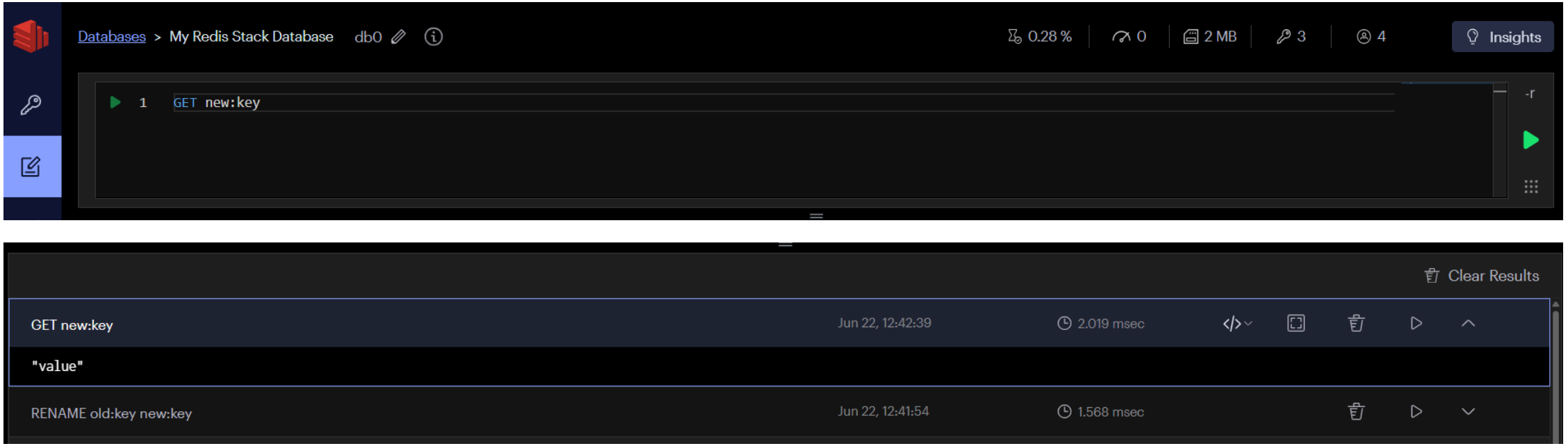
- 키 이름 변경

```
RENAME old:key new:key
```



- 변경된 키 조회

```
GET new:key
```



5. 키 탐색: KEYS와 SCAN

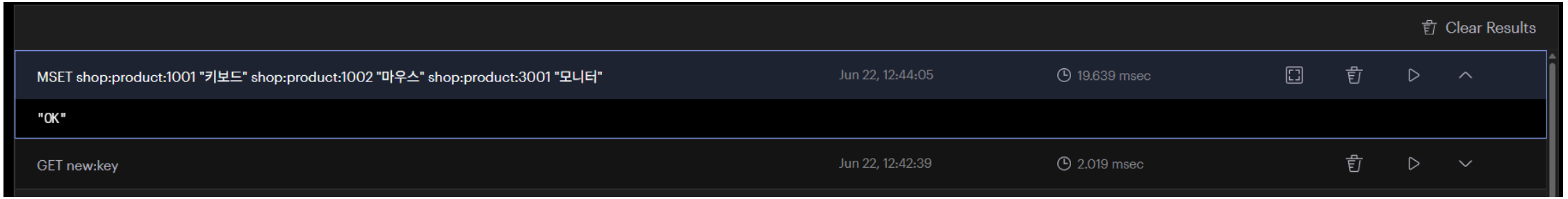
항목	`KEYS`	`SCAN`
목적	조건에 맞는 모든 Key 조회	Key를 조금씩 나누어 조회
반환 방식	한 번에 전체 반환	여러 번 호출하여 순차 반환
성능	O(N) (전체 Key 탐색)	O(1) (호출당 평균)
서버 영향	데이터가 많으면 Redis가 잠시 응답하지 않을 수 있음	서버 부하가 매우 적음

항목	`KEYS`	`SCAN`
운영 환경 사용	권장하지 않음	권장
개발/테스트	사용 가능	사용 가능
결과 개수	모든 Key	일부 Key (Cursor를 이용해 반복 조회)
Cursor 사용	없음	있음 (`0` 이 될 때까지 반복)
Pattern 검색	`KEYS user:*`	`SCAN 0 MATCH user:*`
COUNT 옵션	없음	있음 (`COUNT 100` 등)
대표 사용 사례	개발 중 데이터 확인	운영 서버에서 Key 조회

테스트용 데이터 생성

- `MSET` : 여러 개의 Key-Value를 한 번에 저장하는 명령어(Multiple SET)

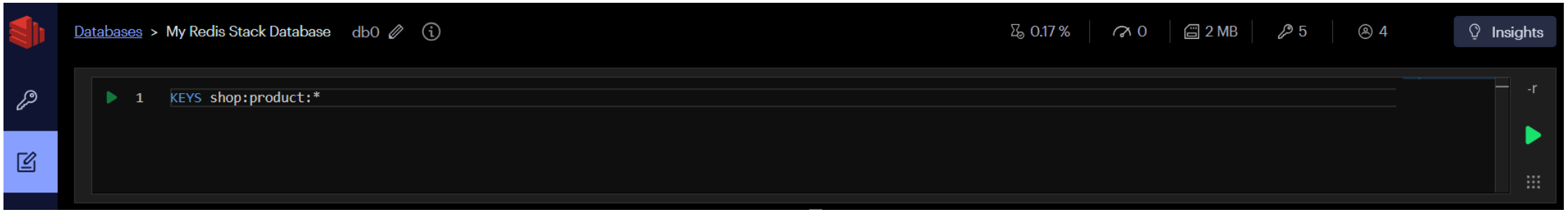
```
MSET shop:product:1001 "키보드" shop:product:1002 "마우스" shop:product:3001 "모니터"
```

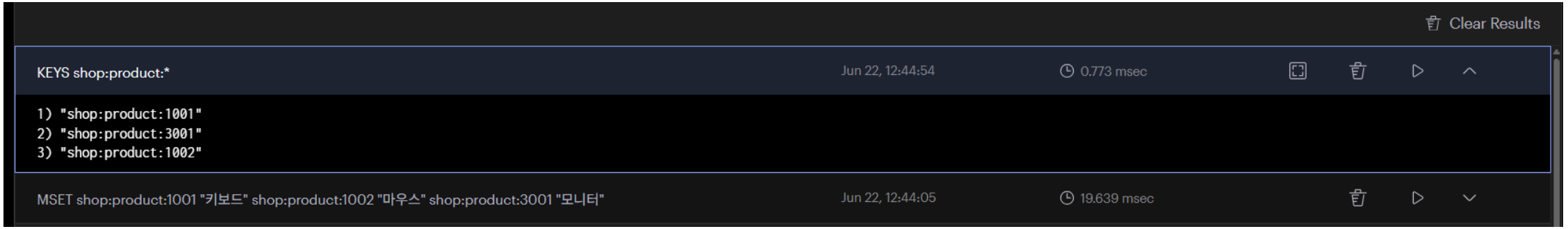


KEYS

- 모든 상품 조회

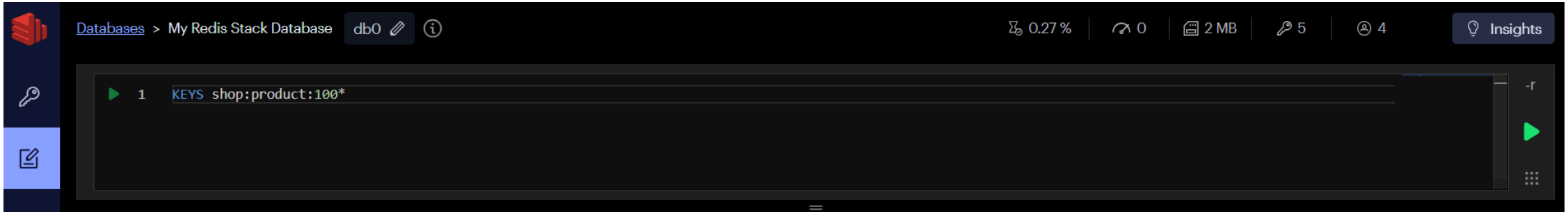
```
KEYS shop:product:*
```

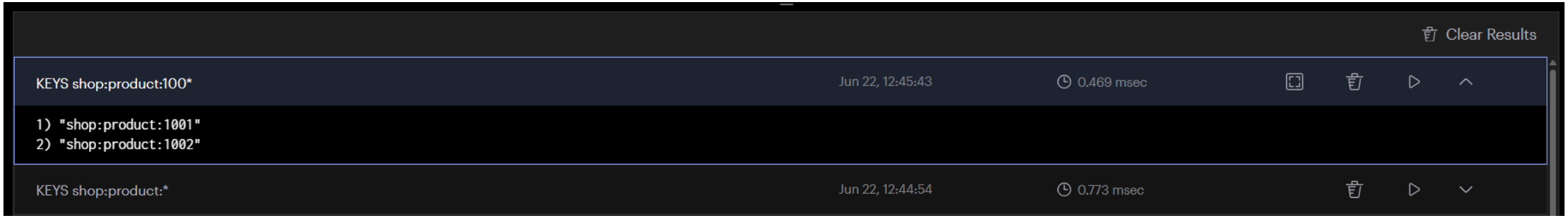




- 특정 상품 조회

```
KEYS shop:product:100*
```

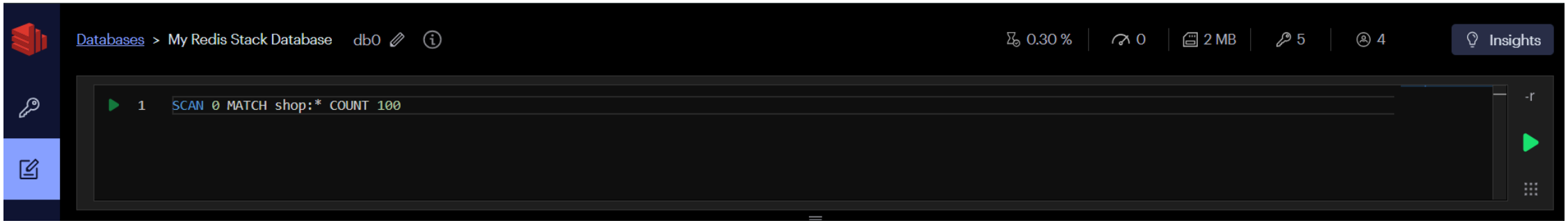


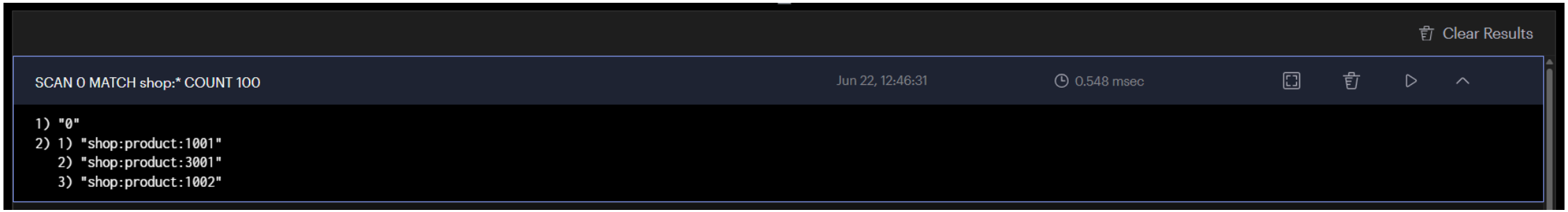


SCAN

- 모든 상품 조회

```
SCAN 0 MATCH shop:* COUNT 100
```





- 특정 상품 조회

```
SCAN 0 MATCH shop:product:100* COUNT 10
```